# ALIGNMENT RESULTS WITH INTERPRETATION

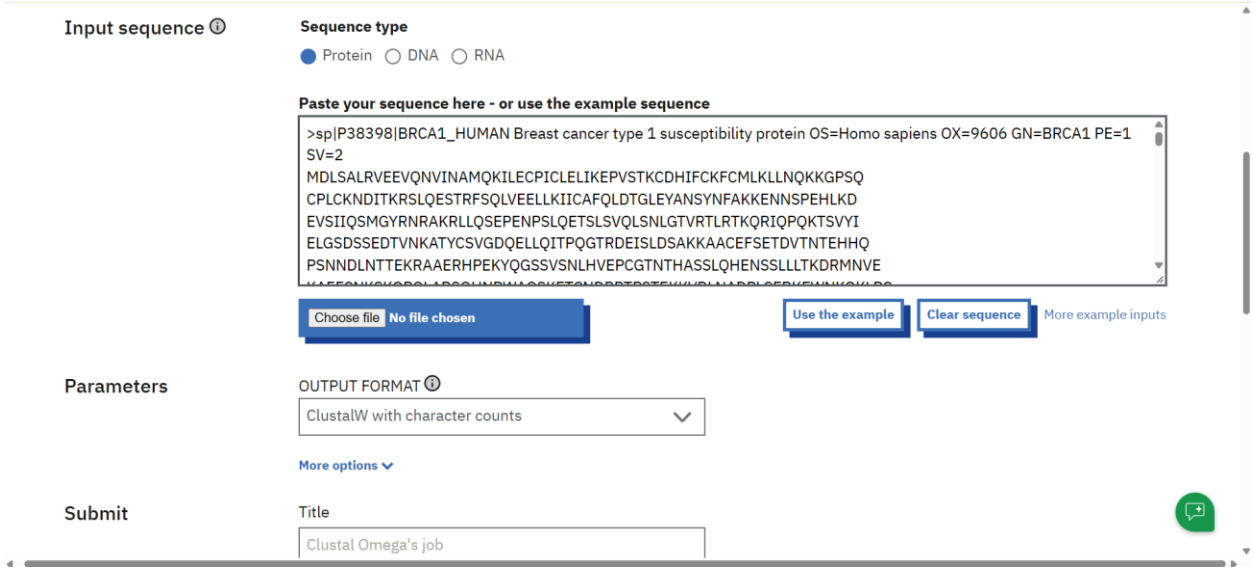



**Fig :** Clustal omega input

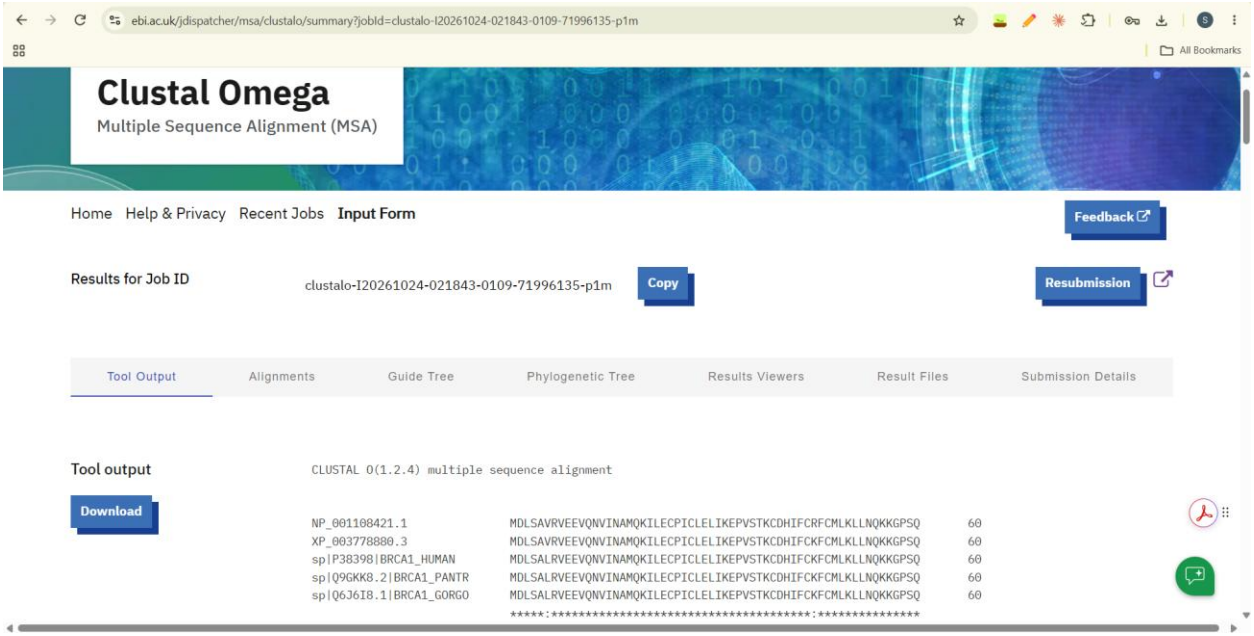



**Fig :** Clustal omega result/header

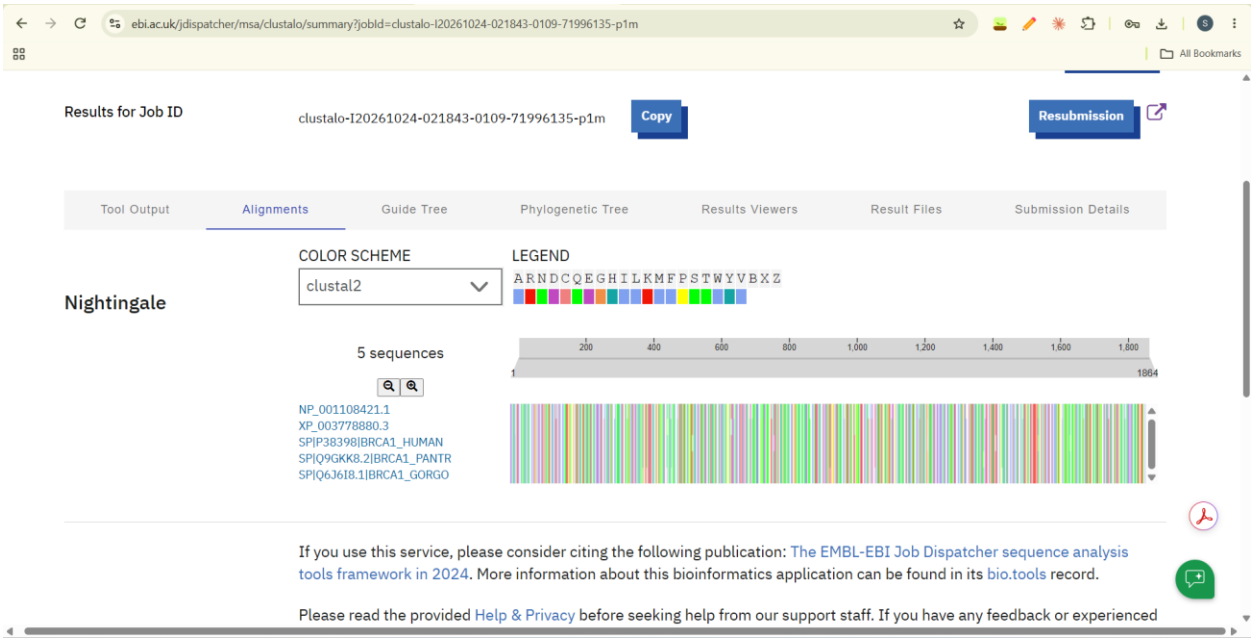



**Fig :** MSA Overview (The MSA shows strong sequence similarity among all five BRCA1 proteins, with many conserved positions across the alignment.)




**Fig :** MSA_conserved_region_1 (This region contains highly conserved amino acids, indicating an important functional role in BRCA1.)

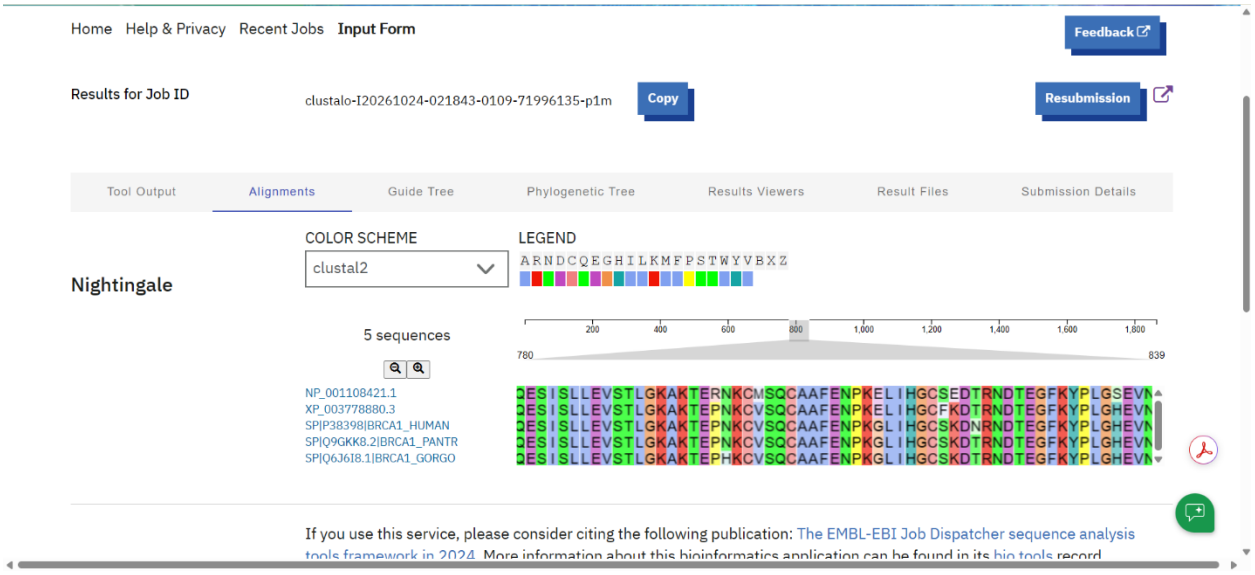



**Fig :** MSA_conserved_region_2 (Strong conservation is observed across the five species, suggesting structural and functional importance.)

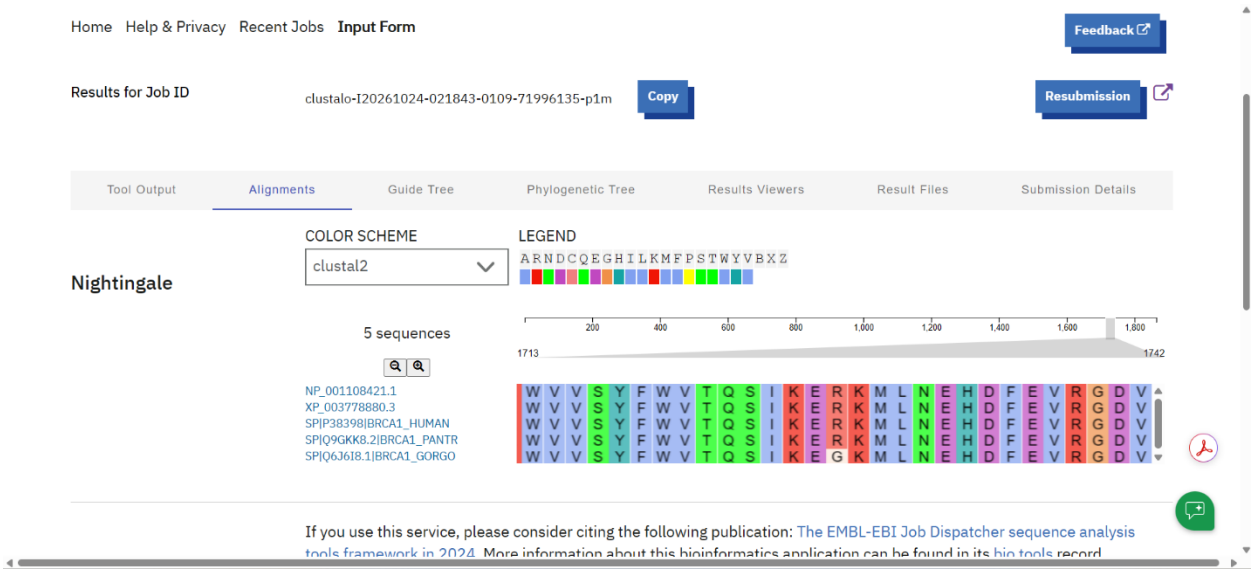



**Fig :** MSA_conserved_region_3 (The high sequence conservation indicates that this region is evolutionarily important and likely contributes to BRCA1 function.)




**Fig :** N-terminal conservation (Positions 1–360)

The N-terminal region is highly conserved, including the characteristic BRCA1 RING-domain residues, supporting its functional importance.

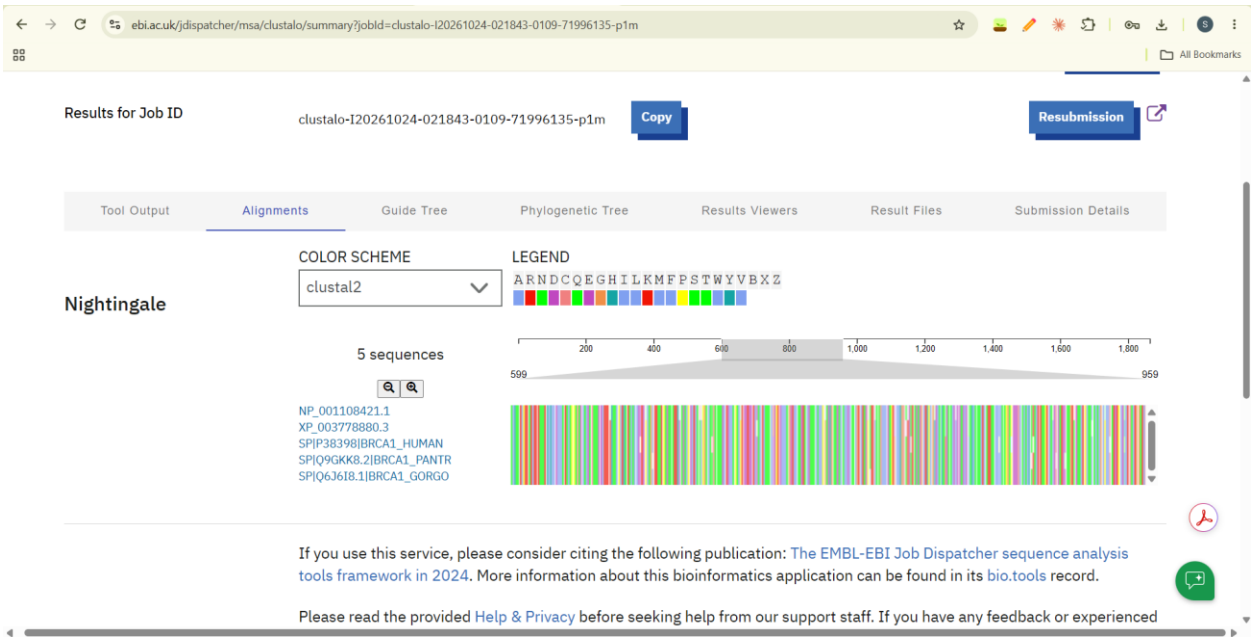



**Fig :** Central/cysteine rich-conservation (Positions 600–960)

This region shows strong conservation, particularly around cysteine-rich residues, suggesting an important structural and functional role.




**Fig :** C-terminal conservation (Positions 1600–1863)

The C-terminal region is highly conserved among the five sequences, indicating strong evolutionary and functional importance.